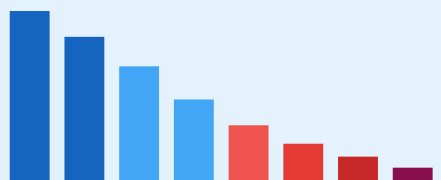


1

PROBLEM

Long-tail Object Detection



LVIS rare classes are under-represented

Semantic prompts help, but add FP risk

Evaluation subsets

4132 GT image-category pairs LVIS
1000 3,217 COCO GT pairs **1000**
LVIS images COCO images

COCO evaluation

3,217 GT pairs **IoU AP50-proxy**

Baseline limitation

- ✗ single primary prompt has limited coverage
- ✗ semantic evidence increases false positives
- ✗ prior-aware fusion is not best overall; it helps LVIS long-tail thresholding

2

METHOD

Semantic Prompt Generation

WordNet

WordNet
semantic variants
and related terms

Templates

attributes · contexts ·
descriptions

Prompt pool per class

airplane

aircraft

vehicle

visible airplane

photo of airplane

OWL-ViT Encoding

Image encoder

Text prompts

Zero-shot object detection → boxes, labels,
scores

Preprocessing

Images resized/normalized for OWL-ViT

Ground-truth category annotations parsed

LVIS + COCO evaluated with recall and AP50-
proxy

3

FUSION CORE

Semantic Fusion Strategies

Max Fusion

Highest prompt evidence score

mean score: 0.3292

Adaptive Fusion

confidence-aware aggregation

mean score: 0.2184

Weighted Fusion

Mean score: 0.1436

Mean Fusion

Mean score: 0.1420

Evidence modules

Multi-prompt evidence coverage

Entropy and semantic conflict modelling

LVIS frequency-aware rescaling

Motivation: recall gain with FP control

4

RESULTS

Key Quantitative Results

Recall improves on both datasets
Category recall

AP50-proxy gains

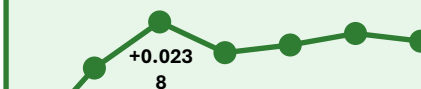
LVIS: 0.2343 → 0.2687
COCO: 0.3687 → 0.4379

False positives

Semantic prompts increase FP counts

LVIS recall: 0.8486 → 0.9143

COCO recall: 0.8901 → 0.9387



LVIS rare AP50-proxy gain

Uncertainty bins show higher FP-rate in high-
risk cases

5

SAFE CONTRIBUTIONS

Research Question Findings

RQ1

Semantic
evidence gives
strongest recall
coverage.

RQ2

Max Fusion has
highest mean score;
Adaptive is more
selective.

RQ3

High
uncertainty/conflict
bins show higher FP
risk.

RQ4

Descriptions provide the
strongest complementary
evidence among evaluated
prompt sources.

RQ5

Prior-aware fusion
improves LVIS adaptive-
threshold recall for rare
categories.

RQ6

Semantic Evidence
achieves the strongest
overall recall coverage
across evaluated
frameworks.No extra training data; diagnostic fusion
analysis for OVD